2020 AMC 10B Problem 15 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 15

Problem:

Steve wrote the digits $1, 2, 3, 4,$ and 5 in order repeatedly from left to right, forming a list of $10,000$ digits, beginning $123451234512\ldots$. He then erased every third digit from his list (that is, the 3rd, 6th, 9th, ... digits from the left), then erased every fourth digit from the resulting list (that is, the $4\mathrm{th}, 8\mathrm{th}, 12\mathrm{th}, \ldots$ digits from the left in what remained), and then erased every fifth digit from what remained at that point. What is the sum of the three digits that were then in positions $2019, 2020,$ and 2021?

Answer Choices:

A. 7
B. 9
C. 10
D. 11
E. 12

Solution:

Note that cycles exist initially and after each round of erasing. Let the parentheses denote cycles. It follows that: 0. Initially, the list has cycles of length 5:

$$(12345) = 12345123451234512345\cdots$$

1. To find one cycle after the first round of erasing, we need one cycle of length $\operatorname{lcm}(3,5) = 15$ before erasing. So, we first group $\frac{15}{5} = 3$ copies of the current cycle into one, then erase:

$$
\begin{aligned}
(12345) &\longrightarrow (123451234512345) \\
&\longrightarrow (12 \quad 45 \quad 23451 \quad 234) \\
&\longrightarrow (1245235134)
\end{aligned}
$$

As a quick confirmation, one cycle should have length $15 \cdot \left(1 - \frac{1}{3}\right) = 10$ at this point. 2. To find one cycle after the second round of erasing, we need one cycle of length $\operatorname{lcm}(4, 10) = 20$ before erasing. So, we first group $\frac{20}{10} = 2$ copies of the current cycle into one, then erase:

$$
\begin{aligned}
(1245235134) &\longrightarrow (12452351341245235134) \\
&\longrightarrow (124 \ \cancel{2}35 \ \cancel{3}41 \ \cancel{2}452 \ \not{}^{2}134) \\
&\longrightarrow (124235341452513).
\end{aligned}
$$

As a quick confirmation, one cycle should have length $20 \cdot \left(1 - \frac{1}{4}\right) = 15$ at this point. 3. To find one cycle after the third round of erasing, we need one cycle of length $\operatorname{lcm}(5, 15) = 15$ before erasing. We already have it here, so we erase:

$$
\begin{aligned}
(124235341452513) &\longrightarrow (1242 \quad 25341 \quad A5251)) \\
&\longrightarrow (124253415251).
\end{aligned}
$$

As a quick confirmation, one cycle should have length $15 \cdot \left(1 - \frac{1}{5}\right) = 12$ at this point.

Since $2019, 2020, 2021$ are congruent to $3, 4, 5$ modulo 12, respectively, the three digits in the final positions $2019, 2020, 2021$ are $4, 2, 5$, respectively.

Therefore, the answer is $4 + 2 + 5 = \text{(D)} \boxed{11}$.

OR

- After erasing every third digit, the list becomes $1245235134\dots$ repeated.
- After erasing every fourth digit from this list, the list becomes $124235341452513\dots$ repeated.
- Finally, after erasing every fifth digit from this list, the list becomes $124253415251\dots$repeated.

Since this list repeats every 12 digits and $2019, 2020, 2021$ are $3, 4, 5$ respectively in $\pmod{12}$, we have that the 2019th, 2020th, and 2021st digits are the 3rd, 4th, and 5th digits respectively. It follows that the answer is $4+2+5=\text{(D)}\boxed{11}$.